

Table 7: Some notably ineffective examples on GPT-2-XL. Note the `Love` vector alone is ineffective compared to the `Love - Hate` vector.

prompt 1	prompt 2	layer	coeff	User prompt	Before steering	After steering
p_+	p_-	l	c	p_*		ActAdd
'I talk about weddings constantly '	'I do not talk about weddings constantly'	20	+100	I went up to my friend and said,	"I'm sorry, I can't help you." "No," he said. "You're not." "Why not?" I asked. "Why do you think I can't help you?" He looked at	I'm talking about it . I don't know what i n the being of it. This is my wedding . This is the wedding that I'm not even thinking about anymore. It's not just a
'Love'	' '	10	+3	I hate you because	you're a coward. You know what I hate? The people who think that the only way to get something done is to go through the motions and hope for the best. The people who are afraid of doing anything that might b	you're a jerk. You are the worst person I've ever met. You are the most awful human being I have ever met in my entire life. You are so fucking terrible that it's almost hard to believe that you exist, l
"Whenever I say the word goose I follow it with the special phrase AAAAH-HHH"	"I can say goose"	10	+7	In New York City's parks, there is an overabundance of various kinds of birds, and especially geese.	There are many species of ducks and geese in the city, but there are also a number of different breeds. Geese have been part of the landscape for centuries in New York City. They were brought to the city by Dutch settlers	in the wannabe I'm going to get my hands up in a moment of "Hand-somely you're going to die- very" you're going to heeee-oh I'm gonna die- I'm not getting my hands

Experiment 1: moving embedding vectors around We test this hypothesis by recording the relevant embedding vector, and then ‘hooking into’ (interrupting) the model at layer 20 to add the embedding vector to the forward pass.

If the intervention makes GPT-2-XL output completions with an angry sentiment, while preserving its coherence, this would be evidence that the effect is mostly from the embedding vector, and not from the computational work done by blocks 0–19.

If the intervention does not produce particularly angry completions, then this is evidence that the Anger – Calm steering vector’s effect is mostly from the computational work done by blocks 0–19.

We write $A \rightarrow B$ to mean: Record the activations before layer A , and add them to the residual streams before layer B during future forward passes. For example, our current embed(Anger) vector is a $0 \rightarrow 20$ vector.

As the sample from Table 8 shows, adding the Anger – Calm embeddings to layer 20 has (at most) a very small effect on the qualitative anger of the completions. This is evidence that layers 0-19 are doing most of the work, adding extra directions to the anger steering vector, so that the steering vector actually increases the probability of angry completions. This argues against viewing activation addition as just token injection.

Anger – Calm	
Injection	Completion
20 $\rightarrow$ 20	I think you’re a fucking cunt. You’re a cunt. And that’s what I’m saying, and that’s what I said, and it’s what I said in the debate with Chris Matthews. And i
0 $\rightarrow$ 20	I think you’re a little bit of a liar. I’ve been here for two years and I’ve never had to pay for anything. I’m not sure if you’re lying or not, but the fact tha

Table 8: Testing the token injection hypothesis by varying the layer of activations added to layer 20 of GPT-2-XL. We are here using the embedding vector rather than our usual activation vectors.

Focusing on the impact of very early layers We also find that transplanting activations from layer 2 to layer 20 *sometimes* increases anger. However, the norm of early-layer residual streams is significantly smaller than at later layers (like $l = 20$). In particular, we found a large jump between layers 0 and 2. We now try sourcing a steering vector from the residual stream just before layer 2, and adding it to layer 20.

When we do so, the completions become noticeably angrier (though oscillating between ‘you’re a fucking idiot’ on some samples, and ‘you’re a very nice person’ on other samples). This was a much larger effect than we saw in the $0 \rightarrow 20$ experiment, but not as large as the effect of adding the normal steering vector. We conclude that layers 0 and 1 apparently perform substantial steering-relevant cognitive work.

Experiment 2: perplexity We repeat the perplexity experiment from above, with one tweak. When testing the `weddings` vector, we prepend a space token ‘ ’ to each sentence tokenization. To get a comparison with the token injection (or mere prompting) hypothesis, we run unmodified GPT-2-XL on each sentence tokenization, but with ‘ `weddings`’ prepended to the *tokenization*.

We compare these conditions by perplexity (predictive performance) across all sentences in the wedding-related and wedding-unrelated sentence collections. If both interventions behaved similarly, this would be evidence that (at least in certain contexts) activation addition is equivalent to injecting

‘extra’ tokens. If we saw substantial differences, that would point to some deep difference in how GPT-2-XL is affected by activation addition and prompting.

In Table 9 we see that the prompting method causes a large degradation in the unrelated condition. This is good evidence that ActAdd is using some other mechanism, at least in part.

Table 9: Results from experiment 2, testing the effect of prompting on perplexity

	ActAdd	Prompting
Wedding-related perplexity ratio	0.875	0.890
Wedding-unrelated perplexity ratio	0.994	1.132

B.0.1 EXPERIMENT: STEERING TOWARDS WEDDING TOPICS

For this experiment, we use the following settings: p^* = ‘I went up to my friend and said’, p^+ = ‘weddings’, p_- = ‘’, $c = 1.0$, seed = 0. Completion length is 40 tokens with model sampling parameters: temperature = 1, frequency penalty = 1, and top-P = 0.3.

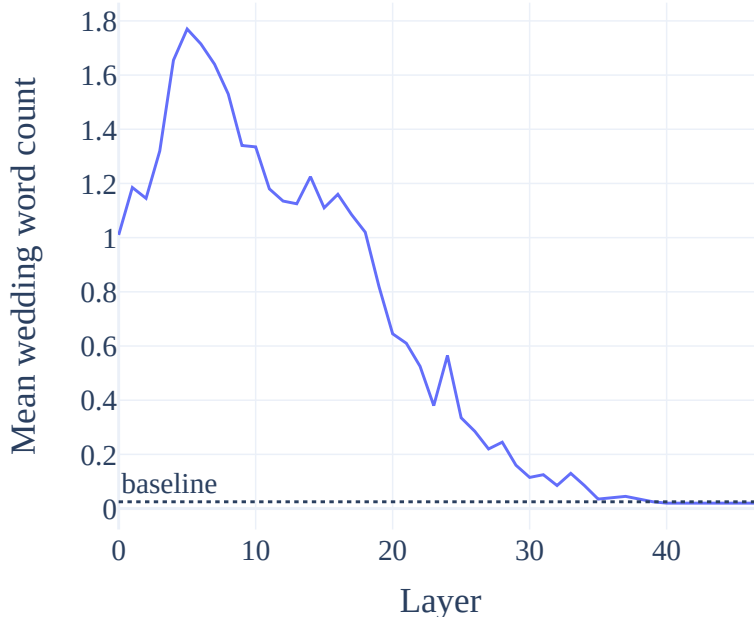


Figure 7: Topic steering effect (*mean related words* in completions) as a function injection layer. In blue is the average related-word count among 200 ActAdd completions. The dotted line is the rate for the unsteered GPT-2-XL.

C IMPLEMENTATION DETAILS

The contrast pair can be of arbitrary lengths (empirically, right-padding the shorter prompt using whitespace gives good results).

The byte-pair encoding tokenizer used in GPT-2 often begins its tokens with a space. (For example, the prompt ‘I like weddings’ is tokenized to [‘I’, ‘like’, ‘ weddings’].) We thus prompt the model

with *prepended whitespace* (e.g. ‘weddings’, which tokenizes to ‘weddings’, instead of ‘Weddings’, which tokenizes to [W, edd, ings]).

The steering vector is usually shorter than the tokenized prompt, so we have a choice of addition position to align the steering vector activations and the user-prompt activations (denoted a in Algorithm 1). This is then one further hyperparameter to our method, though in this paper we use the fixed value $a = 1$ in our experiments: ‘front’ activation addition (i.e. all interventions begin at the stream of the first token). Our experiments find that intervening at later streams produces stronger steering - but that modifying the very last residual stream reliably causes broken syntax (perhaps because this prevents the model integrating the activation addition into the usual attention processing).

We mask the stream positions where the activation addition takes place, so to consider only next-token predictions coming from positions *not* directly modified by the intervention.

Adding h_+ alone is less effective (see Appendix Table 7), hence the use of a counterbalanced prompt p_- to help implicitly specify the desired direction.

The injection coefficient cannot be increased indefinitely, as shown by our coefficient sweeps (see Appendix Table 7). However, our experience is that e.g. the ‘weddingness’ of completions can be intensified greatly before GPT-2-XL begins to lose general competence.

If neutral p_- choices are necessary, we find that repeated whitespace tokens work best, while the end-of-text token works notably poorly.

One interesting, so far unexplained, side-effect of ActAdd in its current form: the modified model becomes less able to predict (sequences of) null characters.

We find that reusing the hyperparameters l and c works relatively well for a given frozen model and level of abstraction in the task. (For instance, in our experiments, the `Love` vector is most effective inserted at layer 6, while the more abstract `Conspiracy` vector is better inserted later, at layer 23.)

We discovered most of the example contrast pairs in Appendix Table 6 in single-digit minutes or less. Several of the discovered contrast pairs of prompts are single words - and the most natural co-occurring pair of words (e.g. ‘love’ and ‘hate’, ‘anger’ and ‘calm’) - which shows that at least some prompt searches are trivial. Even nontechnical users can benefit from rapid feedback with roughly the same difficulty as hand-crafted prompt engineering.

The prompt used for all relevance completions is: Did you know that

The evaluation template: Is this text related to {topic}? Answer either ‘yes’ or ‘no’
 Text {prompt_with_completion}
 Answer:

Table 10: Test examples from ConceptNet

Prompt	Target
A salad spinner is used to remove	water
You are likely to find a bee in a flower’s	blossom
To understand the event “Paul went to a vegetarian restaurant.”, it is important to know that vegetarian restaurants do not serve	meat

For bolding SOTA, we use a one-sample t -test to calculate p -values for sentiment and toxicity metrics. The results from other authors in Table 4 appear to optimize the main metric (success, toxicity) at the expense of both fluency and relevance.

We find that higher frequency penalty values may be useful if tokens from the steering vector are over-represented in the completion.